

Management Information Systems

Phase 2: Relational Database Architecture & Analytical SQL Implementation

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1 Comprehensive Data Architecture & Schema Definition

The enterprise relational schema is engineered to capture the full operational lifecycle of copper extraction, from field surveys and comminution to inventory tracking and quality assaying. Below is the structural and logical breakdown of each relation:

1.1 1. Teams Relation (Teams)

The **Teams** table models the organizational and human resource backbone of plant operations, enabling operational task assignment and personnel traceability:

- **TeamID (Primary Key):** Unique identifier referenced as a Foreign Key across operational and sampling tables.
- **TeamName & LeadName:** Identifies operational squads and the supervising engineer accountable for shift performance.
- **Location & Shift:** Enforces strict operational constraints. Every team is stationed in a designated municipal/mine jurisdiction and operates within an explicit shift cycle (Morning, Afternoon, or Night).
- **Field:** Encapsulates technical domain specialization (e.g., Comminution, Quality Sampling). Intersheet referential integrity ensures teams are strictly dispatched to workshops and tasks matching their certified field and municipality.

1.2 2. Geological Surveys Relation (GeologicalSurveys)

Records exploration borehole drilling coordinates, stratigraphic evaluations, and preliminary mineral grade indicators. Spatial attributes strictly align with corresponding downstream processing locations, such as crushing facilities and regional storage hubs.

1.3 3. Equipment Assets Relation (Equipment)

Manages the mechanical inventory of the mining plant. Records unique machine IDs, equipment categories (e.g., Jaw Crushers, Ball Mills, Conveyors), models, procurement dates, maintenance timestamps, and operational status:

- **Status:** Restricted via domain validation to three enumerations:
 - **Operational**
 - **Maintenance**
 - **Out of Service**
- **Operational Dispatch Logic:** Downstream execution units (such as **CrushingOperations**) are restricted to selecting assets tagged strictly as **Operational**. Furthermore, each asset is physically constrained to a single municipal jurisdiction (e.g., Equipment unit #141 cannot operate across multiple distinct cities).

1.4 4. Comminution Operations Relation (CrushingOperations)

Captures primary rock fragmentation records. Each row represents a single crushing shift executed by a designated operational team utilizing an active machinery unit at a specific site:

- **Operational Integrity:** Strict checks guarantee that all three daily operational shifts operate with localized teams and active machines.
- **Mass Conservation Formula:** Material throughput adheres to physical mass conservation:

$$\text{Output Weight} = \text{Input Weight} - \text{Waste Weight} \quad (1)$$

1.5 5. Energy Analytics Relation (EnergyConsumption)

Acts as a 1-to-1 analytical extension to **CrushingOperations**, logging precise energy draw and power tariffs:

- **Attributes:** **Energy_Consumed(KWh)**, **Electricity Cost per Unit (\$/kWh)**, and **Total Energy Cost (USD)**.
- **Governing Equation:**

$$\text{Total Energy Cost (USD)} = \text{Energy_Consumed (kWh)} \times \text{Electricity Cost per Unit (\$/kWh)} \quad (2)$$

1.6 6. Intra-City Haulage Fleet Relation (Trucks)

Manages haul truck cycles transporting crushed ore from comminution discharge hoppers to local processing warehouses. The model enforces an intra-city transit boundary constraint: trucks exclusively transfer materials between facilities within the same municipality.

1.7 7. Warehouse Infrastructure Relation (Warehouse)

Models the physical distribution depots and storage bins:

- **WarehouseID (Primary Key):** Foreign Key anchor for **Inventory**.
- **Capacity Utilization Metric:** Measures volumetric stock levels relative to engineered design capacity:

$$\text{Current_Utilization_Percent} = \frac{\sum \text{CurrentVolume}_{\text{Warehouse}}}{\text{CapacityTons}} \times 100\% \quad (3)$$

An index exceeding 100% triggers hazardous overfill alarms; low percentages indicate under-utilized buffer capacity.

- **Storage_Cost_each_month(USD):** Dynamically calculated proportional to active storage occupancy.

1.8 8. Mineral Inventory Relation (Inventory)

The central material register tracking lot movements per storage depot:

- **Dynamic Stock Balance Formula:**

$$\text{CurrentVolume} = \text{Weight_In} - \text{Weight_Out} - \text{Waste} \quad (4)$$

- **MaterialType:** Categorizes stockpiles (Run-of-Mine ore, Flotation Slurry, Filter Cake, Refined Cathodes) to drive dispatch scheduling.
- **Upstream Traceability:** Each record forms an anchor for quality audits in **Sampling**.

1.9 9. Quality Assurance Sampling Relation (Sampling)

Logs lot sample extractions linked via **InventoryID**. Captures sampling timestamps, total gross sample weight, specimen count, sampling margin of error, and **TeamID** responsible for collection. Functions as the audit bridge between static warehouse stock and laboratory testing.

1.10 10. Laboratory Quality Control Relation (QualityControl)

Logs quantitative chemical assay determinations performed on samples identified via **SampleID**:

- **Dry Sample Mass Determination:** Eliminates moisture bias from grade evaluations:

$$\text{Dry_Weight} = (1 - \text{Moisture_Rate}) \times \text{SampleWeightKg} \quad (5)$$

- **Composite Quality Score (Test_Score):** A multivariate linear index evaluating sample quality by rewarding copper purity and penalizing deleterious elements (Iron, Zinc) and moisture:

$$\begin{aligned} \text{Test_Score} = & 0.5 \times \text{Purity_Percent} + 0.3 \times \text{Copper_Percent} \\ & - 0.1 \times \text{Iron_Percent} - 0.1 \times \text{Zinc_Percent} - 0.2 \times \text{Moisture_Rate} \end{aligned} \quad (6)$$

2 Analytical SQL Query Formulation & Technical Logic

Query 1: Energy Consumption & Total Cost Aggregated by Municipality

Objective: Calculate the total comminution electrical power consumption and cumulative expenditure partitioned by municipality/geographic location.

Tables Involved: `CrushingOperations`, `EnergyConsumption`

Analytical Utility: Highlights energy-intensive regional extraction nodes to support municipal energy negotiations and regional cost optimization.

Query 2: Quality Assays & Mean Copper Purity Grouped by Material Category

Objective: Evaluate the metallurgical grade of stockpiled mineral classes (ROM ore, concentrate, refined metal) by computing mean purity percentages and composite test scores.

Tables Involved: `Inventory`, `Sampling`, `QualityControl`

Analytical Utility: Distinguishes premium-grade lots suitable for immediate commercial shipment from lower-grade batches requiring additional leaching or blending.

Query 3: Warehouse Capacity Utilization & Overfill Risk Audit

Objective: Identify storage facilities operating beyond certified nameplate design capacity or exhibiting severe underutilization.

Tables Involved: `Warehouse`, `Inventory`

Analytical Utility: Directs logistics teams to rebalance bulk stock transfers, averting safety violations and storage bottleneck hazards.

Query 4: Comminution Circuit Scrap Ratio & Waste Generation Rate

Objective: Compute the percentage of waste produced during crushing:

$$\text{Waste Rate (\%)} = \left(\frac{\text{Waste Weight}}{\text{Input Weight}} \right) \times 100$$

Tables Involved: `CrushingOperations`

Analytical Utility: Identifies shifts and crushing units suffering from anomalous rock wear, uncrushable tramp material, or sub-optimal closed-side settings.

Query 5: Moisture Content Comparative Analysis (High vs. Low Grade Samples)

Objective: Segment samples by quality threshold (`Test_Score` ≥ 50 vs. < 50) and compare their average moisture content.

Tables Involved: `QualityControl`

Analytical Finding: Demonstrates that moisture percentages remain virtually uniform between high and low-scoring samples, proving that elemental impurities (Iron, Zinc) rather than water content drive grade divergence.

Query 6: Sampling Team Evaluation via Laboratory Assay Quality

Objective: Evaluate sampling squad precision by computing the mean `Test_Score` of samples gathered by each operational team.

Tables Involved: `Teams`, `Sampling`, `QualityControl`

Analytical Utility: Detects potential sampling bias, field contamination, or procedural non-compliance attributable to specific field crews.

Query 7: Active Inventory Stock Balance Partitioned by Material Type

Objective: Display current on-hand stockpile volumes segregated by mineral classification per warehouse facility, ordered ascending by WarehouseID.

Tables Involved: Inventory, Warehouse

Analytical Utility: Provides immediate enterprise-level stock visibility for production planning and sales order dispatching.

Query 8: Unit Energy Expenditure per Ton of Comminuted Ore

Objective: Compute mean electrical cost per metric ton of processed input ore across each operational shift:

$$\text{Unit Cost} = \frac{\text{Total Energy Cost}}{\text{Input Weight}}$$

Results are sorted in descending order to surface the least cost-efficient operations.

Tables Involved: CrushingOperations, EnergyConsumption

Query 9: Monthly Warehousing Expenditure Aggregated by Municipal Region

Objective: Sum cumulative monthly inventory holding costs across all operating storage facilities categorized by municipal region.

Tables Involved: Warehouse

Analytical Utility: Essential input for logistics capital budgeting and comparative regional cost analysis.

Query 10: Top 5 Storage Facilities Exhibiting Severe Inventory Spoilage

Objective: Identify the five warehouses reporting the largest cumulative inventory waste tonnage, sorted descending.

Tables Involved: Inventory, Warehouse

Analytical Utility: Pinpoints warehouses suffering from compromised storage conditions, improper ventilation, or material degradation during handling.

Query 11: Quantitative Assay of Absolute Dry Weight and Contained Elemental Copper

Objective: Join multi-table attributes to compute dry net sample weights and calculate the actual contained elemental copper mass:

$$\text{Pure Elemental Cu Mass} = \text{Dry_Weight} \times \text{Copper_Percent}$$

Correlates these values with material type and host storage warehouse.

Tables Involved: QualityControl, Sampling, Inventory, Warehouse

Query 12: Multi-Dimensional Quality Assay Profile by Depot and Material Class

Objective: Multi-table aggregation calculating the Minimum, Maximum, and Mean `Test_Score` for every material category across distinct warehouses, filtering for locations with multiple tested lots and sorting descending by average score.

Tables Involved: `QualityControl`, `Sampling`, `Inventory`, `Warehouse`

Analytical Utility: Provides a comprehensive statistical profile to benchmark inter-warehouse quality stability.